One shared block is a 16-step schedule, not an iterable operator

Stage S's frozen d320 weights, applied T = 4..24 times instead of 16. No training: T is the only thing that changes.

T 51904636 +51905401 | scored 51919538 | commit aec838b | frozen d320 weights, 3 seeds | 18 grids, 14 rho x 400 | no training

A tied cell has a dial no untied cell has

- Applications T are an eval-time knob on FIXED weights, not retraining
- An operator tolerates T>16; a 16-step schedule peaks at 16
- 18 grids, six T values, three seeds, no training at all

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 393 TOKENS

HINTS	`BEGIN_HINTS N1 COLOR_2 N3 COLOR_1 N4 COLOR_3 N6 COLOR_3 N7 COLOR_3 N8 COLOR_0 ... (59 tokens)`
GRAPH	`BEGIN_GRAPH N0 N47 N3 N26 N33 N43 N1 N23 N32 N11 N12 N38 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 N17 COLOR_1 N23 ... (45 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 5 rounds, 19 deductions,
38 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000011, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

29 hints of 48 nodes, then 5 rounds of simultaneous deduction. Hints: N1=COLOR_2 N3=COLOR_1 N4=COLOR_3 N6=COLOR_3 N7=COLOR_3 N8=COLOR_0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 ...
  11 deduced
BEGIN_ROUND          N18 COLOR_0 N25 COLOR_2 N35 COLOR_1 N47 COLOR_3
  4 deduced
BEGIN_ROUND          N0 COLOR_2 N19 COLOR_0
  2 deduced
BEGIN_ROUND          N46 COLOR_2
  1 deduced
... 1 further round

SOLVED
```

WHY ONE DEDUCTION FIRES

node N5, neighbours N2 N15 N9 N13 N36

their colours known before this round:

```
   [0, 1, 2]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N2
```

only colour left: 3

```
so the trace emits  N5 COLOR_3
```

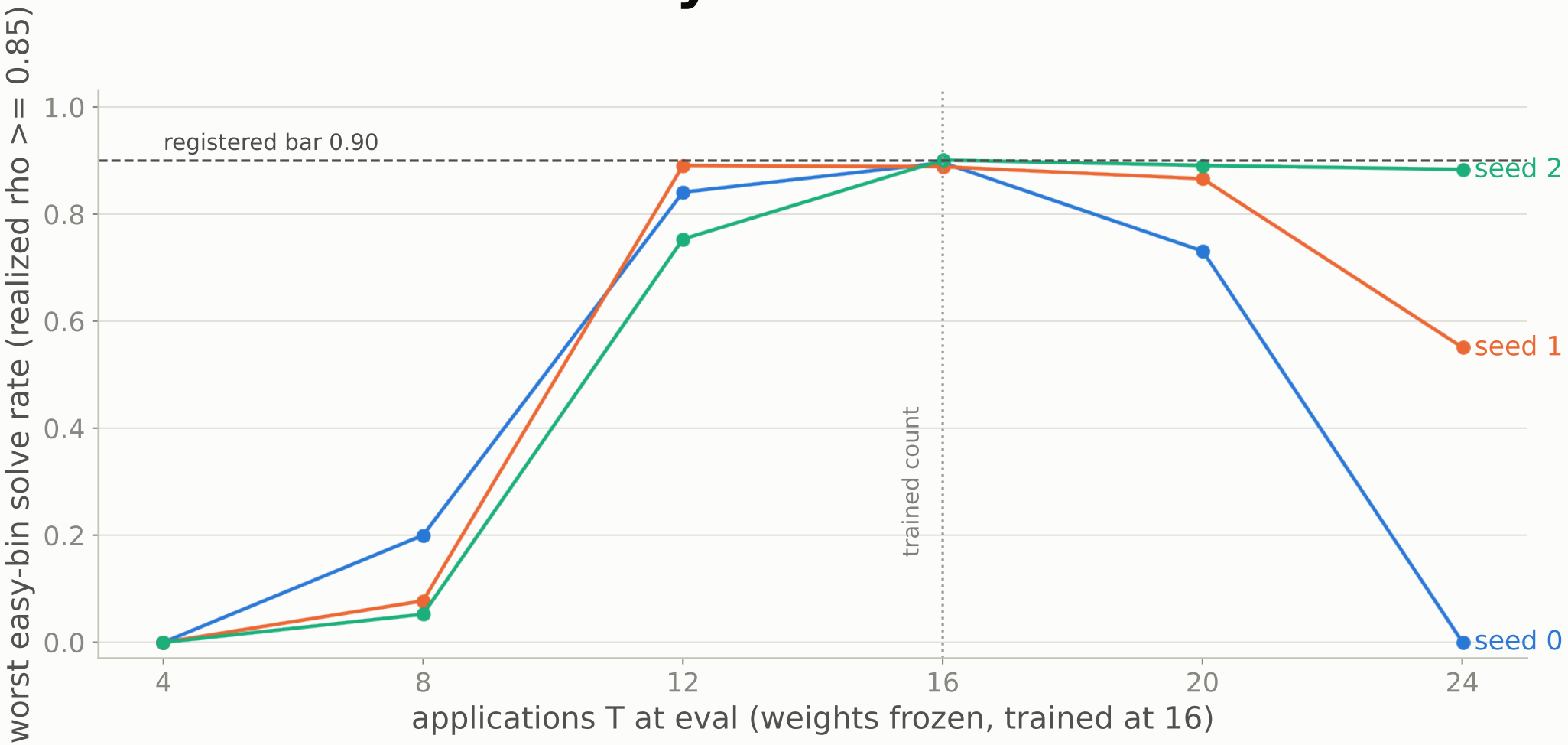
The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the same distribution stage S trained on, re-scored here at six values of T.

T=5 rounds, the median of 40 draws from this config (range 1-19) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

No T clears the easy-end bar in all three seeds



T1, read first because it cannot censor. At the trained count one seed sits exactly on the bar and the worst misses by 0.0125.

At T=24 the near-miss becomes a collapse: one seed hits zero

T	worst easy bin, per seed	worst	vs bar 0.90
T=4	0.000 / 0.000 / 0.000	0.0000	-0.9000
T=8	0.200 / 0.077 / 0.052	0.0525	-0.8475
T=12	0.840 / 0.890 / 0.752	0.7525	-0.1475
T=16 (trained)	**0.895 / 0.887 / 0.900**	**0.8875**	**-0.0125**
T=20	0.730 / 0.865 / 0.890	0.7300	-0.1700
T=24	0.000 / 0.550 / 0.882	0.0000	-0.9000

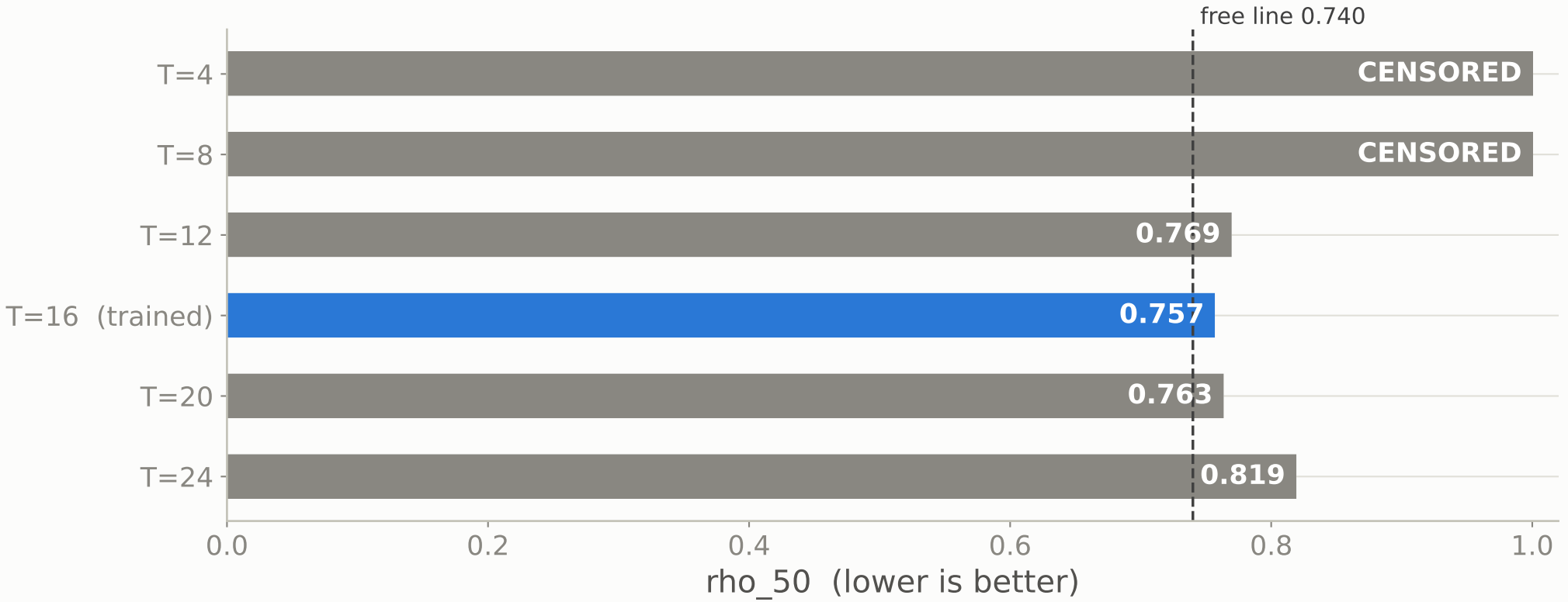
Per seed, not a median: at T=24 the three seeds are 0.000, 0.550 and 0.882, and their median describes none of them.

rho_50 bottoms exactly at the trained count, and censors below T=12

T	rho_50	seeds crossing	mean solve	vs T=16
T=4	CENSORED	0/3	0.0000	-0.3148
T=8	CENSORED	0/3	0.0159	-0.2989
T=12	0.769	3/3	0.2963	-0.0186
T=16 (trained)	**0.757**	**3/3**	**0.3148**	**--**
T=20	0.763	3/3	0.3020	-0.0129
T=24	0.819	2/3	0.1873	-0.1275

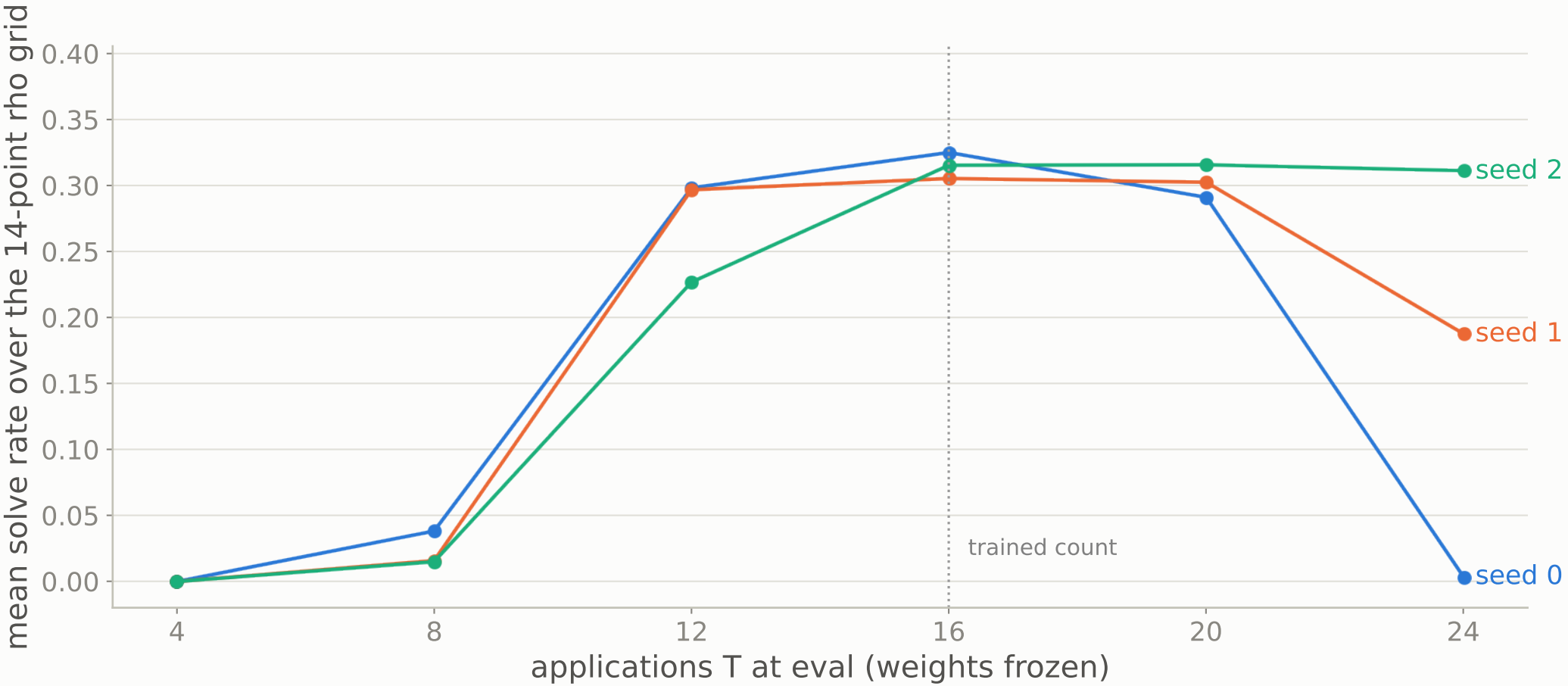
CENSORED means no density bin cleared a 0.5 solve rate, so there is no rho_50 to quote. Lower rho_50 is better.

No T beats the trained count, and T=24 is 0.062 worse



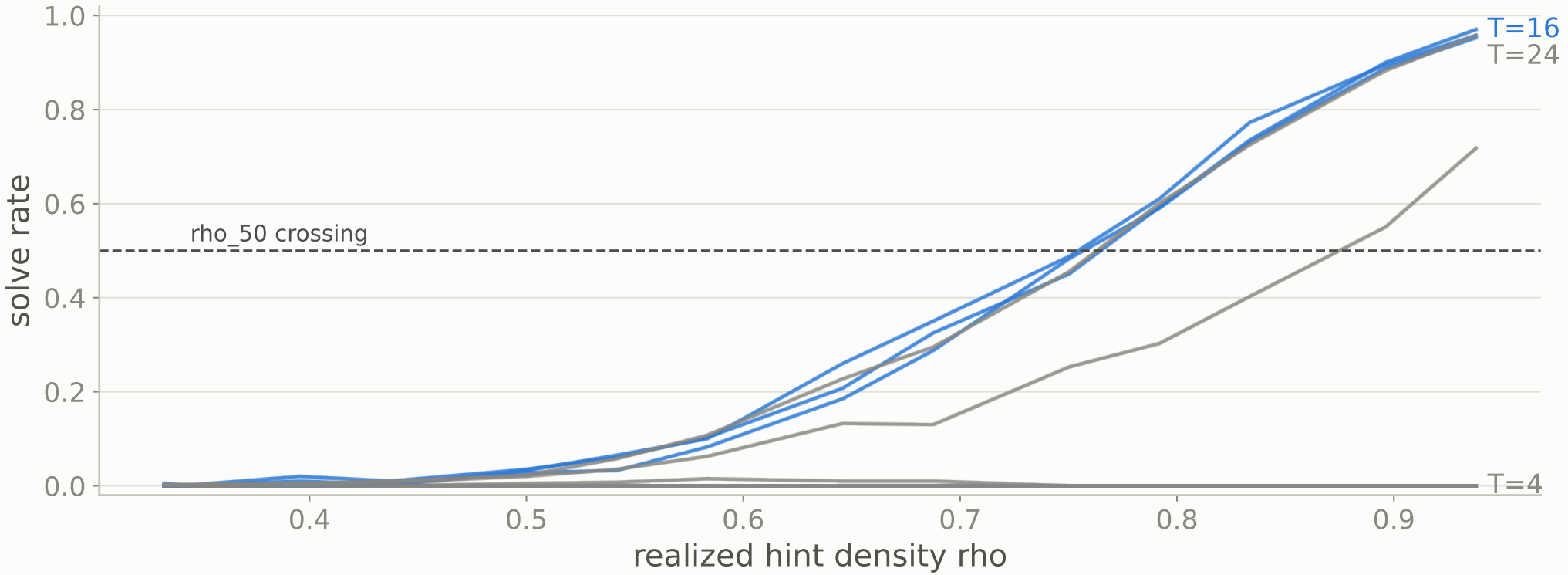
T0 was registered to hold if every T >= 16 landed within 0.01 of the best. The spread above 16 is 0.062.

No seed improves past T=20; 2 of 3 peak at 16



Seed 2 is flat out to T=20 (0.3152 against 0.3148). The median over seeds peaks at 16. Nothing was trained.

Too few applications censor; too many degrade the same curve



Three seeds per T. At T=4 no density reaches the dashed line at all; at T=24 one seed is flat on the floor.

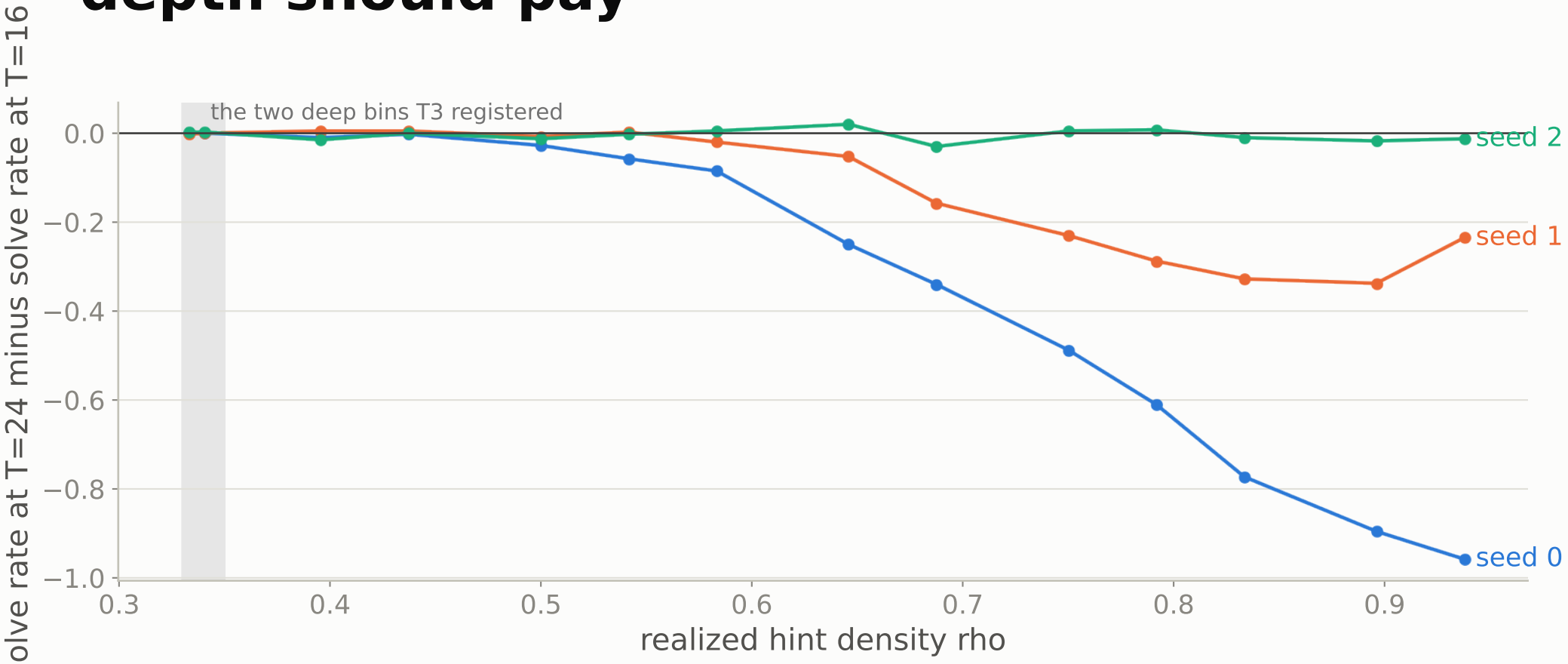
T0 falsified: the trained count is a peak, not a floor

- |T24 - T16| is 0.062 against the registered 0.01 bar
- Both directions fall away; the median mean solve peaks there
- A clean negative, registered in advance as equally publishable

All sixteen applications are used: none is padding

- T* = 16, the smallest T within 0.01 of T=16's mean solve
- T=12 already costs 0.0186 of mean solve against T=16
- So the analysable object does not shrink below sixteen rounds

Over-iterating costs at the easy end, not where depth should pay



T3 registered a positive paired delta at the two lowest-rho bins. It is +-0.005 there in 1 of 3 seeds: nothing.

Stage T speaks only for d320's frozen weights at N=48

- One width, one trained count: no tied cell trained at another T
- Fixed weights, so the only spread is the three trained seeds
- N=48 only; the N=96 bars are a different size (finding 55)

Against the previous best, the T sweep changes nothing: 0.757 stands

quantity	J2 untied d192 (standing)	S1 tied d320, best T
rho_50 median	**0.580**	**0.757**
best T over the sweep	n/a (depth is its weights)	T=16
mean solve	--	0.3148
trainable	2,504,890	613,818
distinct blocks	16	1
seeds crossing 0.5	3/3	3/3

Same 14-point grid, same 400 instances per bin, same --shuffle-adj --fall. No improvement is claimed: no T beats 0.757.

The next cut is k distinct blocks, and it is already running

- Stage U: k blocks cycled T/k times, still one iterated map
- Array 51922710, six cells, k in {2,4} by three seeds
- If no k clears 0.740 the next cut is optimization, not capacity